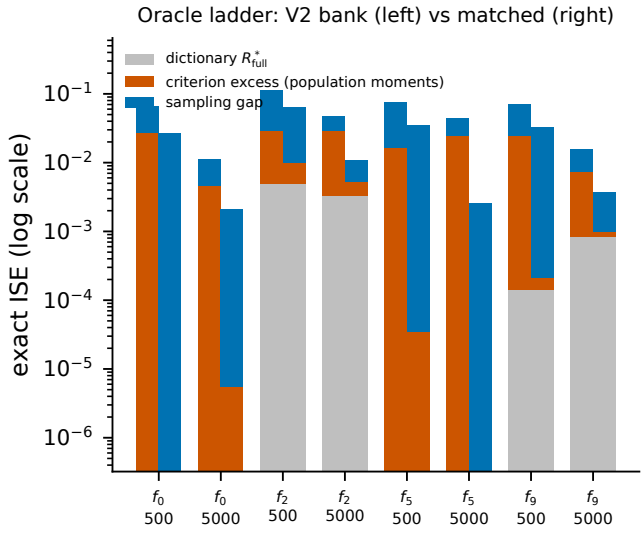
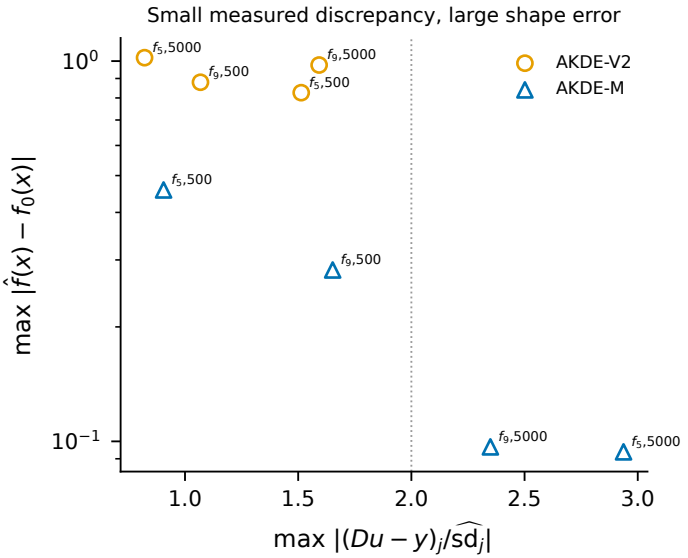
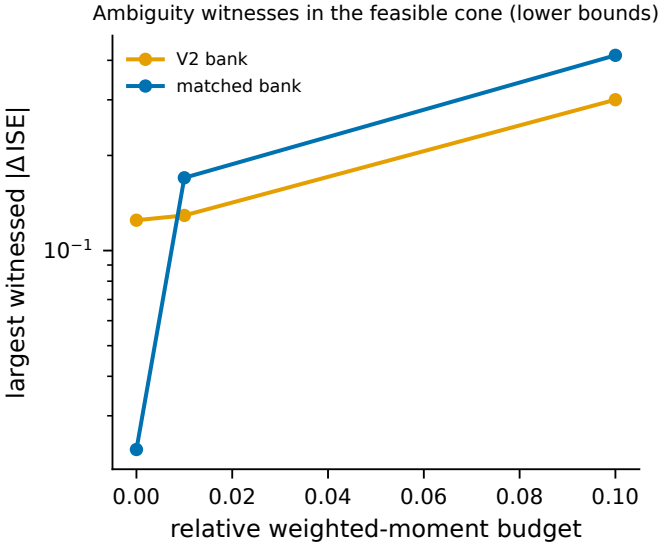
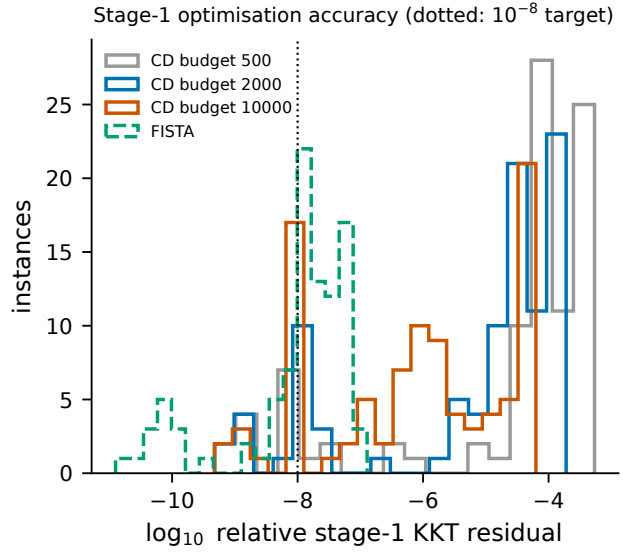

Oracle ladder: V2 bank (left) vs matched (right)
dictionary R^*_{full}
criterion excess (population moments)
sampling gap
exact ISE (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
f_0 500
f_0 5000
f_2 500
f_2 5000
f_5 500
f_5 5000
f_9 500
f_9 5000


Small measured discrepancy, large shape error
AKDE-V2
AKDE-M
f_5,5000
f_9,5000
f_9,500
f_5,500
f_5,500
f_9,500
f_9,5000
f_5,5000
max $|\hat{f}(x) - f_0(x)|$
10^0
10^{-1}
1.0
1.5
2.0
2.5
3.0
max $|(Du - y)_j / \widehat{\text{sd}}_j|$


Ambiguity witnesses in the feasible cone (lower bounds)
V2 bank
matched bank
largest witnessed $|\Delta \text{ISE}|$
10^{-1}
0.00
0.02
0.04
0.06
0.08
0.10
relative weighted-moment budget


Stage-1 optimisation accuracy (dotted: 10^{-8} target)
CD budget 500
CD budget 2000
CD budget 10000
FISTA
instances
0
5
10
15
20
25
−10
−8
−6
−4
$\log_{10}$ relative stage-1 KKT residual